



200W Wideband EMC Benchtop Power Amplifier 0.6GHz-6GHz

Features

- Wideband EMC Solid State Power Amplifier
- Small Signal Gain 60dB Typical
- Output Saturation Power 53dBm Typical
- Supply Voltage 110/220 VAC
- 50 Ohm Matched Input/Output
- Over Temperature Protection
- Over Current Protection
- Auto Calibration

Typical Applications

- Microwave Radio and VSAT.
- Telecom Infrastructure.

Parameter		Min.	Typ.	Max.	Units
Frequency Range		0.6		6.0	GHz
Small Signal Gain		50	60		dB
Gain Flatness			± 5.0		dB
Gain Variation Over Temperature (0°C to +50°C)			± 3.0		dB
Input VSWR			1.4	2.0	:1
Output 1dB Compression Point (P1dB)			49		dBm
Saturated Output Power (Psat) (CW)		52	53		dBm
Supply Current (220V AC)			5	10	A
IM3			-30		dBc
Power Added Efficiency (PAE)			30		%
Turn On/Off Speed (Switch Disable)	ON	200			ns
	OFF	200			ns
Turn On/Off Speed (Drain Disable)	ON	1000			us
	OFF	40			us
Turn On/Off Speed (Gate Disable)	ON	1000			us
	OFF	40			us

Weight	1680 Max. Ounces	Impedance	50ohms
Input / Output Connectors	N-Female	Material	Aluminum
Finish	Dark Grey Baking Paint	Package Sealing	5U Rack-mount/Tabletop Chassis



Absolute Ratings

Supply Voltage	90VAC to 264VAC
RF Input Power (RFIN)	Psat – Large Signal Gain

Biasing Up Procedure

Step 1	Connect input and output with 50 Ohm source/load. (In band VSWR < 1.9:1 or >10dB return loss.)
Step 2	Connect Power Cable
Step 3	Turn On Back Panel AC Power Supply Switch
Step 4	Press Front Panel Power Switch to Power Display

Power OFF Procedure

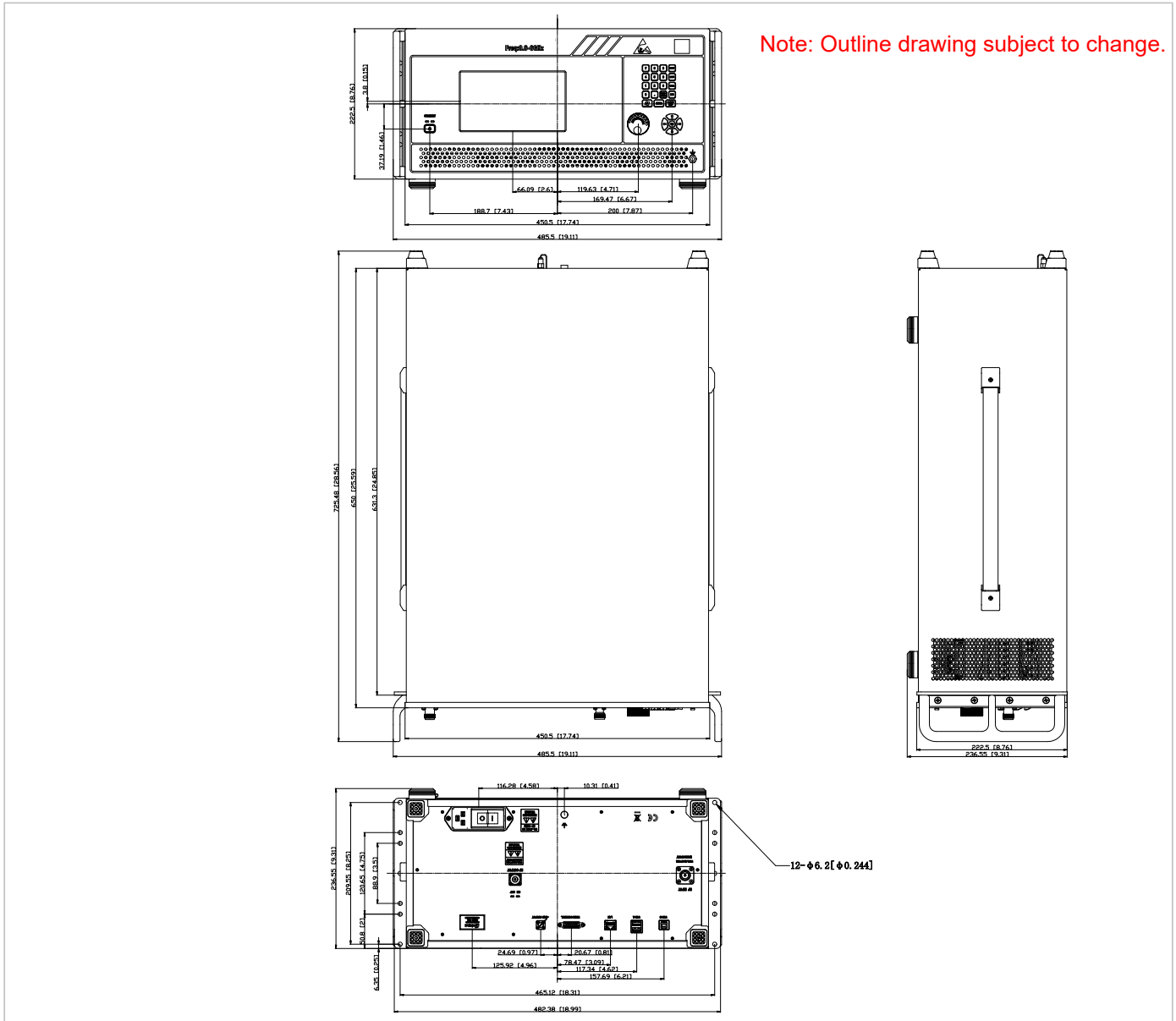
Step 1	Press Front Panel Power Switch to Power Off Display
Step 2	Turn Off Back Panel AC Power Supply Switch
Step 3	Remove Power Cable (If Moving Equipment)
Step 4	Disconnect input and output with 50 Ohm source/load. (In band VSWR < 1.9:1 or >10dB return loss.)

Environmental Specifications

Operational Temperature	0°C~+50°C(Case Temperature)
Storage Temperature	-40°C to +85°C
Altitude	30,000 ft. (Epoxy Sealed Controlled environment)
	60,000 ft. 1.0psi min (Hermetically Sealed Un-controlled environment) (Optional)
Thermal Shock	0°C → +50°C (5 Cycles / 10 hours)
Vibration	25g RMS (15 degrees 2KHz) endurance, 1 hour per axis
High Temperature Burn In	Temperature +50°C for 72 Hours
Humidity	100% RH at 35°C, 95%RH at 40°C
Shock	20G for 11msec half sine wave, 3 axis both directions

Outline Drawing:

All Dimensions in mm (inches)



Packing List

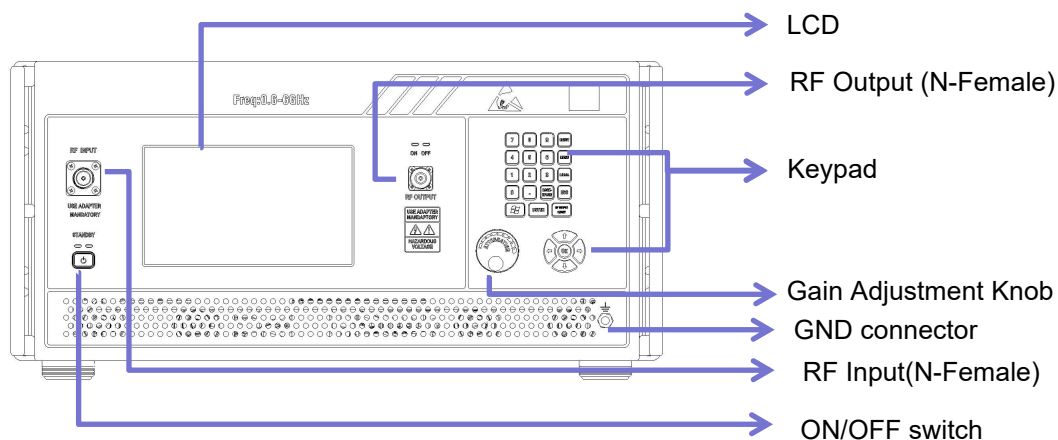
ID	Description	QTY
1	Fig a. AC power supply line (Consulting sales)	1



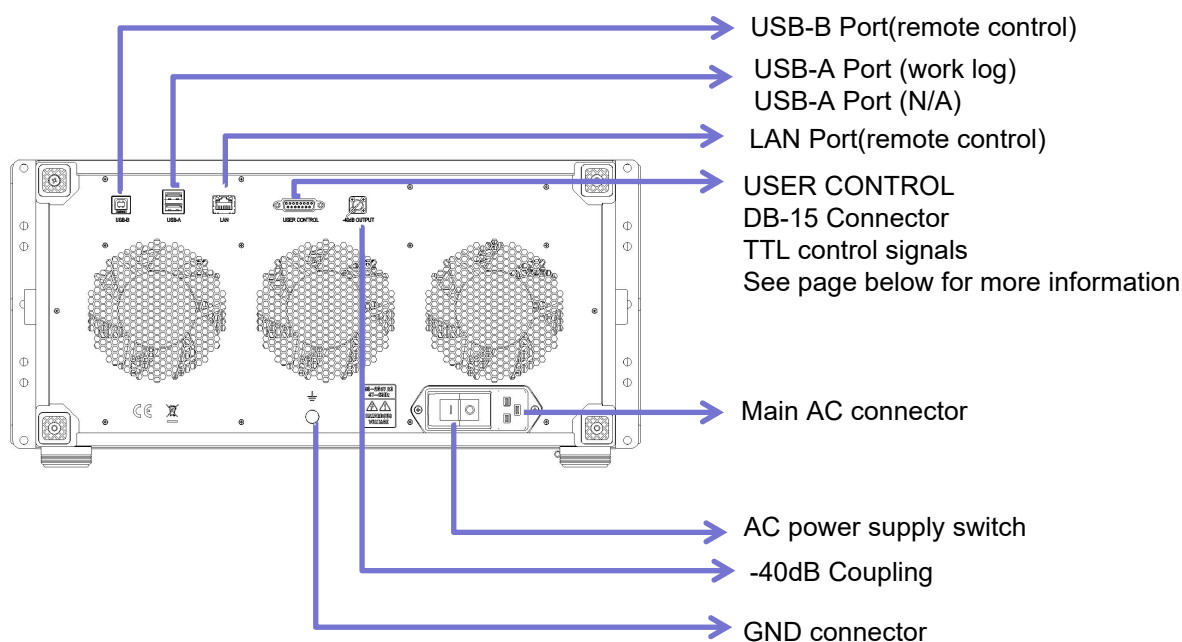
Fig a.

EMC Equipment User Manual

Front Panel



Rear Panel



Front Panel LCD Screen Display

Switching On Instrument



Please follow the instructions on the front panel LCD screen after switching on the power.
Press “1” or “ENTER” on keypad to continue.

Self Calibration Screen

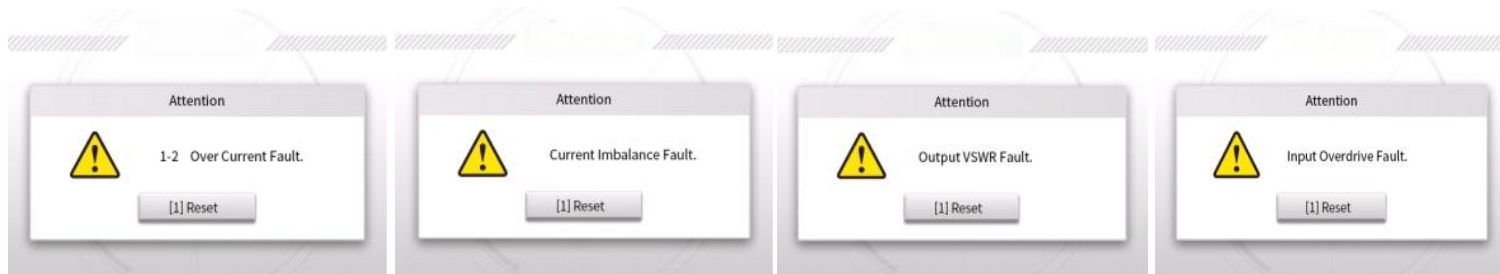


Calibration is may be recommended.
“[1] **Calibrate**” to execute instrument self calibration process.

“[2] **Reset**” to reboot the instrument.

*Please connect input and output with 50 Ohm source/load.

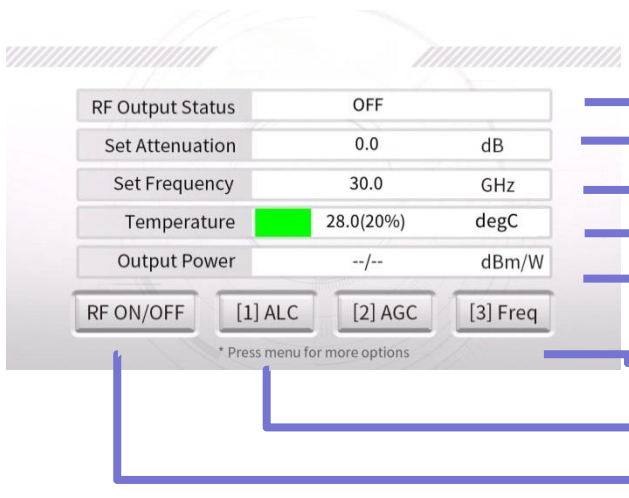
Instrument Protection Alarms



The front panel LCD screen will display the error code or error message when instrument self protection is triggered. Front panel alarm indicator will light up.

To eliminate the error code, press “RESET” on front panel keypad to reboot the instrument and clear the alarms.

Instrument Status Display Page



Indicates instrument RF output status. It will display: ON or OFF.

RF output attenuation (change with adjustment knob).

RF input signal frequency.

Instrument temperature.

Instrument RF output power.

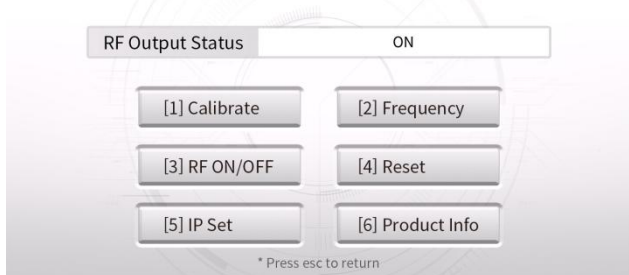
Press "Menu" on keypad to enter instrument functions selection menu.

ALC mode, power lock.

Switches On or Off for instrument RF output port.

Instrument Function Selection Page

To enter this function selection page, press "Menu" on front panel keypad while the instrument is showing the status page. Press the corresponding number on front panel keypad to select:



[1] Calibrate the calibrate key is valid.

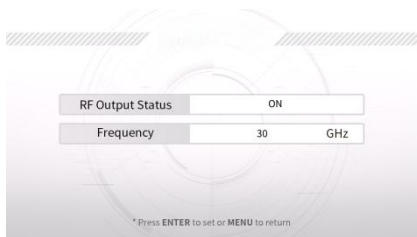
[2] Frequency enters RF input signal frequency.

[3] RF ON/OFF switches the RF output port on or off.

[4] Reset Restarts the instrument (Turns RF output off)

[5] IP Set enter IP display page.

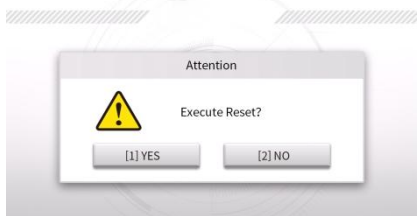
[6] Product Info displays product part number and serial number



RF Output Status: ON

Frequency: 30 GHz

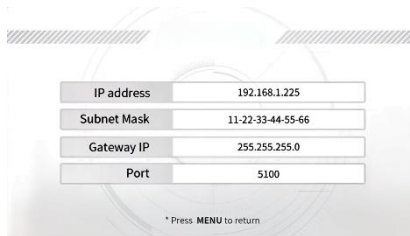
* Press ENTER to set or MENU to return



Attention

Execute Reset?

[1] YES [2] NO



IP address: 192.168.1.225

Subnet Mask: 11-22-33-44-55-66

Gateway IP: 255.255.255.0

Port: 5100

* Press MENU to return



Part Number: DBEMCA26504000B

Frequency Range: 26.5-40GHZ

Serial Number: 123456789

Firmware Version: V2.03

Technical Support: www.qotana.com

* Press MENU to return

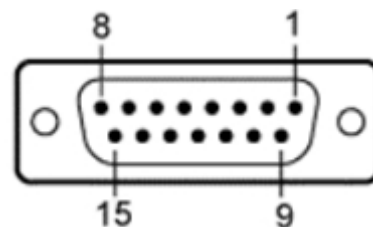
Some functions will ask for confirming execution when selected from function selection menu.

Note: The data shown here is just for presentation.

Protection Connector Table

Female D-Sub is on the housing

The mating male part number: 172-E15-103R001

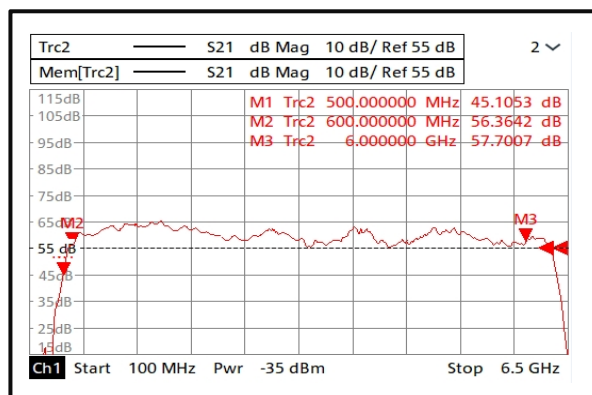


Pin#	Name	Function	Initial State	Description	Applied
1	Reset	Control	HIGH	Resets PA when logic <u>LOW</u> is applied for five more seconds and released	Yes
2	Gate Disable	Control	LOW	Applying logic <u>HIGH</u> disable gates of amplifiers	Yes
3	Drain Disable	Control	LOW	Applying logic <u>HIGH</u> disable drains of amplifiers	Yes
4	RF Input Over Drive	Indicator	LOW	Pin will be latched to logic HIGH when input signal is over limit	Yes
5	Temp Over	Indicator	LOW	Pin will be latched to logic HIGH when amplifier is driven over temperature	Yes
6	Current Over	Indicator	LOW	Pin will be latched to logic HIGH when drain current limit is reached	Yes
7	Current Imbalance	Indicator	LOW	Pin will be latched to logic HIGH when an imbalance in the drain current of the combining branches occurs	Yes
8	PA Off Alarm	Indicator	LOW	Pin will be latched to logic HIGH when any of the protection limit is reached	Yes
9	Fan Alarm	Indicator	LOW	Pin will be latched to logic <u>HIGH</u> when Fan limit is reached	Yes
10	RF Input Switch	Control	LOW	Applying logic HIGH turns OFF RF front-end switch to terminator	Yes
11	VSWR	Indicator	LOW	Pin will be latched to logic HIGH when output reflection is over limit	Yes
12	Fixed Attenuation 10dB	Control	LOW	Applying and holding logic HIGH to enable 10dB fixed attenuation	Yes
13	Fixed Attenuation 20dB	Control	LOW	Applying and holding logic HIGH to enable 20dB fixed attenuation	Yes
14	+5V	Power Supply	+5V	+5V DC is supplied for reference (200mA)	Yes
15	GND	Ground	GND	Ground	Yes

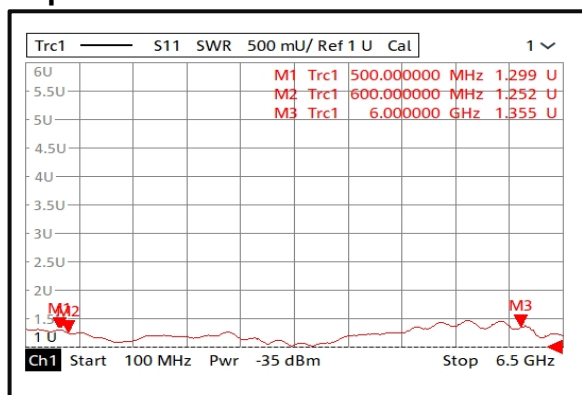
Notes:

- HIGH/LOW voltages are standard TTL signals 0.0V-0.8V = LOW. 2.8V-5V = HIGH.

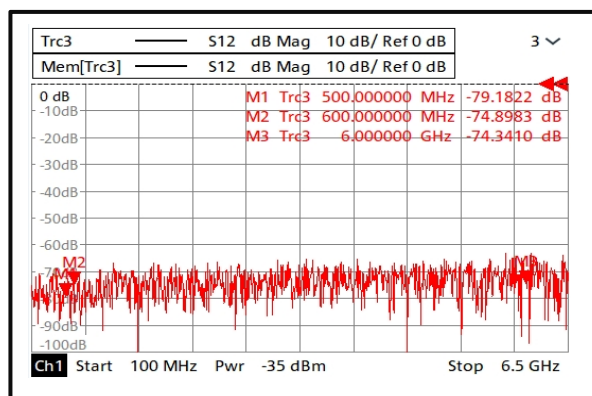
Gain



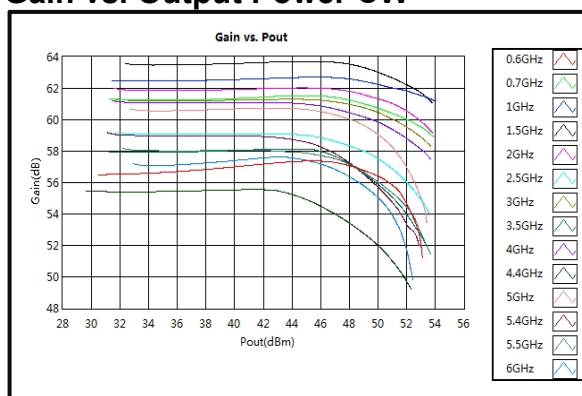
Input VSWR



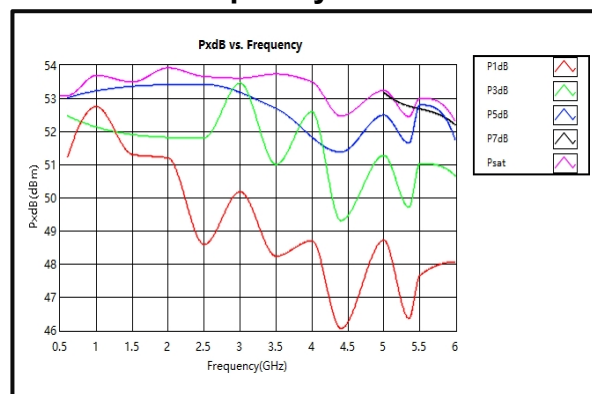
Isolation



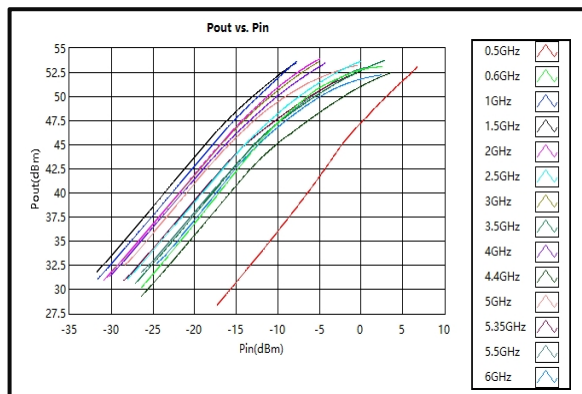
Gain vs. Output Power CW



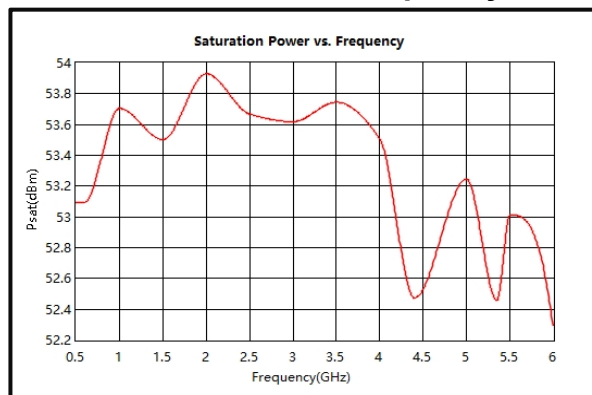
PndB vs. Frequency CW



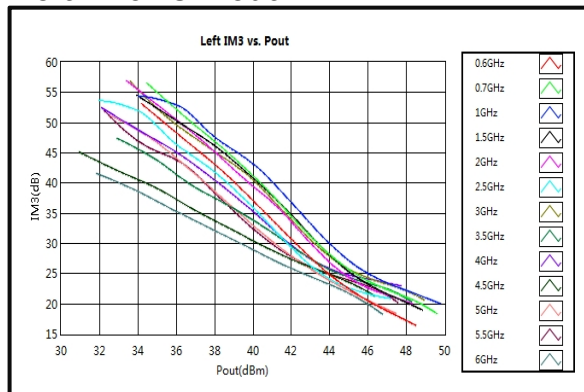
Pout vs. Pin



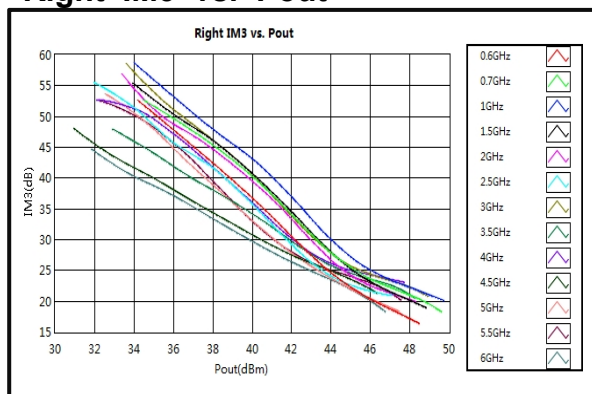
Saturation Power vs. Frequency CW



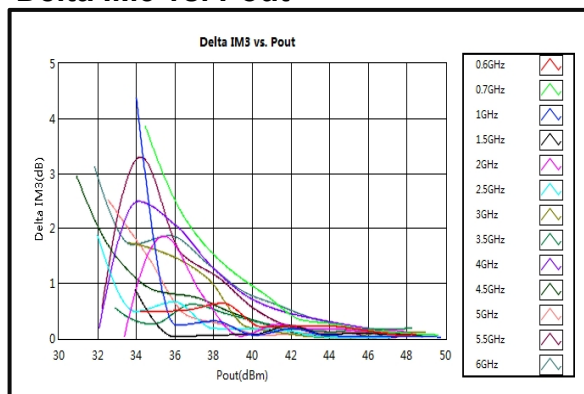
Left IM3 vs. Pout



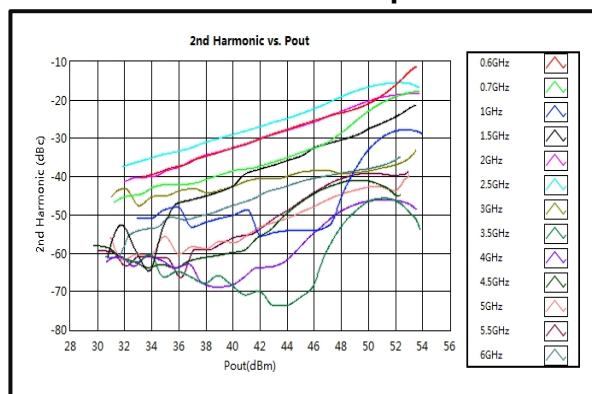
Right IM3 vs. Pout



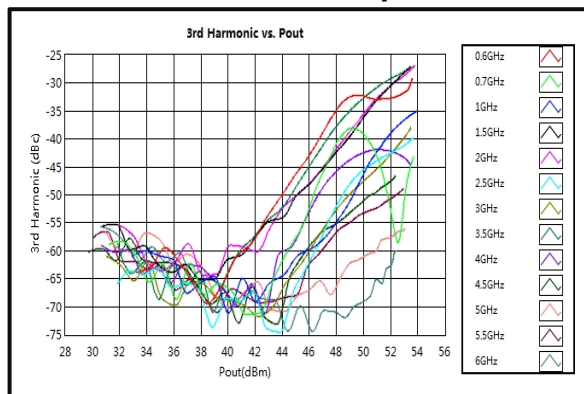
Delta IM3 vs. Pout



2nd Harmonic Wave Output Power

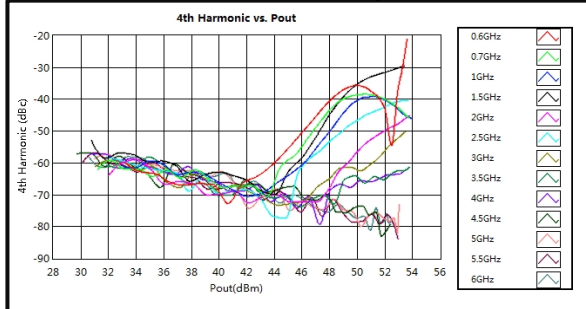


3rd Harmonic Wave Output Power

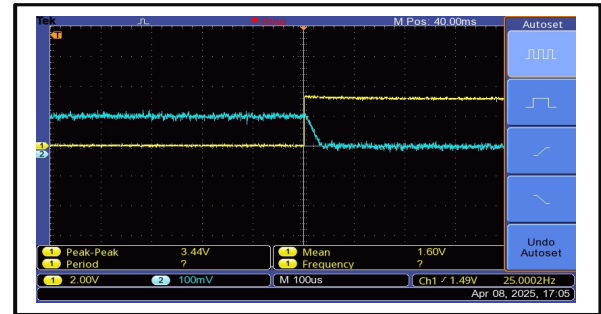




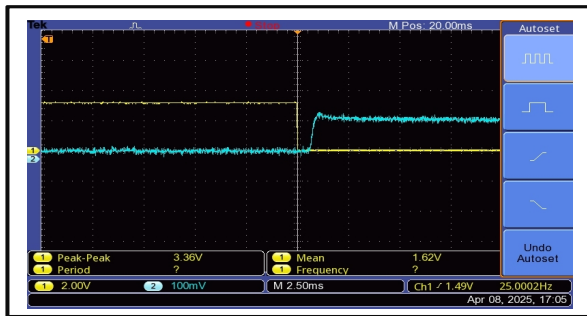
4th Harmonic Wave Output Power



The Gates Closure Time is 40 us @+25°C

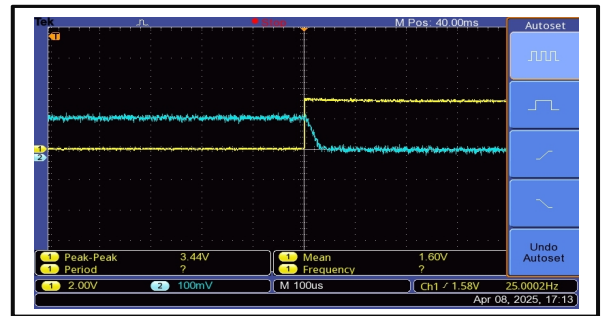


The Gates Open Time is 1000 us @+25°C

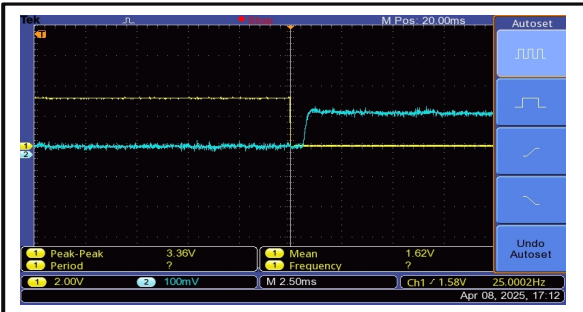


The gate control port: D-sub 15 PIN #2 (Gate Disable).
The yellow curve is the gate control signal, the blue curve is RF output envelope.

The Drains Closure Time is 40 us @+25°C

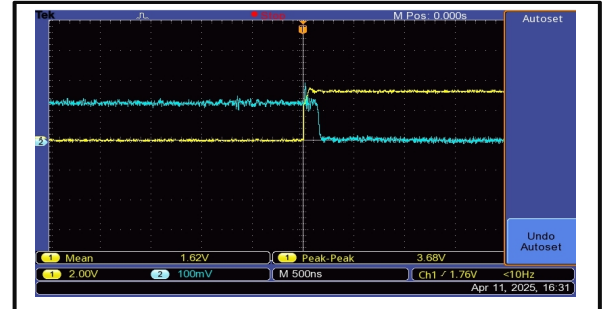


The Drains Open Time is 1000 us @+25°C

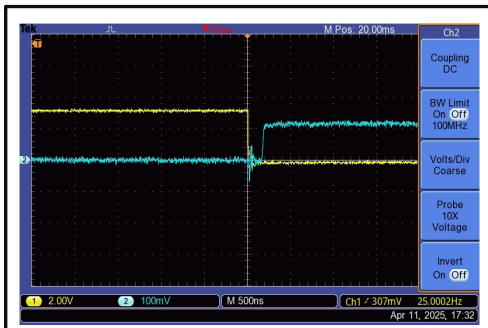


The drain control port: D-sub 15 PIN #3(Drain Disable).
The yellow curve is the drain control signal, the blue curve is RF output envelope.

The RF Switch Off Time is 200 ns @+25°C



The RF Switch On Time is 200 ns @+25°C



Switch control port: D-sub 15 PIN #10(Switch Disable).
The yellow curve is the switch control signal, the blue curve is RF output envelope.